

Student Information

Full Name	Karugu Festus Wanjau
Reg No	C026-01-0731/2023
Submitted	2026-07-28 20:09:40

Overall Result

25/28 (89.3%) Grade: B

Score by Section

Section	Score	Pct
Activation Functions	2/5	40.0%
Weight Initialization	3/3	100.0%
Loss Functions	3/3	100.0%
Optimizers	4/4	100.0%
Backpropagation	3/3	100.0%
Regularization	4/4	100.0%
Batch Normalization	2/2	100.0%
Architecture Design	4/4	100.0%

Questions to Review

Q01. [Activation Functions]

What is the output range of the Sigmoid activation function?
Sigmoid = 1/(1+e^{-z}). As z->+inf output->1, as z->-inf output->0. It never reaches 0 or 1, so the range is the open interval (0, 1).

Q02. [Activation Functions]

Which activation function suffers from the 'Dying ReLU' problem?
ReLU outputs 0 for all z<=0 and its gradient is also 0 there. A neuron stuck at 0 stops receiving gradient updates and dies. Leaky ReLU fixes this with a small slope (0.01) for negative inputs.

Q03. [Activation Functions]

Why is Tanh preferred over Sigmoid for hidden layers?
Tanh outputs range (-1, 1) centered around 0. Sigmoid outputs (0, 1) are always positive, causing zig-zag gradient updates. Zero-centering avoids this inefficiency.

